

Software Ethics

| *"With great power comes great responsibility." — Uncle Ben*

Why building software is never a value-neutral act: guiding engineering decisions to protect human wellbeing, digital rights, and public trust.

The Fundamental Ethical Question

"Does this system do what it should, for whom it should, without harm?"

Code is never neutral — every architectural and algorithmic choice encodes human values, assumptions, and societal consequences.



Moral Judgment Over Algorithms

Ethical codes are not simple automated rules; they require active assessment of tradeoffs and consequences.



Public Welfare Priority

Engineers carry an absolute duty to prioritize societal safety and rights over corporate deadline pressure.



Preventing Negligence

Ignoring known safety risks or rushing defective code directly endangers human life and livelihoods.

Key Dimensions of Software Ethics



1. Data Privacy

Informed consent & minimal data collection.



2. Bias & Fairness

Auditing systems against systemic discrimination.



3. Accountability

Human ownership of system outcomes.



4. Transparency

Auditable decision paths for users.



5. Autonomy

Eliminating deceptive dark patterns.



6. Security Duty

Addressing negligent flaws as ethical failures.



7. Sustainability

Minimizing carbon & compute footprints.



8. IP & Authorship

Fair licensing in AI generative models.

Privacy & Data Ethics vs. Bias & Fairness

DIMENSION 01

Privacy & Data Ethics

Consent must be informed, explicit, and unbureled. Gather only necessary data.

CASE STUDY: CAMBRIDGE ANALYTICA (2018)

Facebook data of 87M users harvested via quiz apps for political psychographics without explicit consent. Resulted in a \$5B FTC fine.

DIMENSION 02

Algorithmic Bias & Fairness

Algorithms encode historical human biases unless proactively tested and mitigated.

CASE STUDY: AMAZON AI RECRUITING (2018)

Internal hiring tool trained on 10-year male-dominated resumes penalized candidates with the word "women's". Scrapped before deployment.

Accountability vs. Transparency & Explainability

DIMENSION 03

System Accountability

"The algorithm did it" is not an ethical or legal defense.
Humans must own code outcomes.

CASE STUDY: VW DIESELGATE & BOEING MCAS

VW deliberately coded defeat software (11M cars affected). Boeing 737 MAX MCAS software concealed risks, causing 346 fatal crashes.

DIMENSION 04

Transparency & Explainability

Opaque "black-box" systems prevent users from contesting automated life decisions.

CASE STUDY: DUTCH CHILDCARE BENEFITS SCANDAL (2021)

An unexplainable risk algorithm falsely flagged thousands of dual-national families as fraud suspects, bankrupting victims and dissolving the government.

User Autonomy vs. Security as an Ethical Duty

DIMENSION 05

Autonomy vs. Dark Patterns

Software UX should support user intention, not manipulate cognitive vulnerabilities for revenue.

CASE STUDY: FTC V. AMAZON "ILIAD FLOW" (2023)

Amazon sued for designing a labyrinthine multi-step cancellation flow ("Iliad") to prevent users from unsubscribing from Prime while signup took one click.

DIMENSION 06

Security as an Ethical Duty

Negligent security patching and software race conditions inflict direct physical and financial harm.

CASE STUDY: THERAC-25 & EQUIFAX BREACH

Therac-25 software race conditions fatal-dosed radiation patients. Equifax exposed 147M records by ignoring a known Apache Struts patch for months.

Environmental Sustainability & IP / Generative AI

DIMENSION 07

Environmental Sustainability

Computation carries immense environmental costs in energy, water cooling, and carbon output.

CASE STUDY: LLM TRAINING CARBON FOOTPRINTS

Strubell et al. (2019) demonstrated training a single large model emits CO2 equal to 5 cars over their lifetime, spurring the Green AI movement.

DIMENSION 08

IP & Generative AI Authorship

Generative AI blurs lines between data training reuse, licensing compliance, and original authorship.

CASE STUDY: NYT V. OPENAI & GITHUB COPILOT

Landmark lawsuits challenging unpermitted scraping of copyrighted news articles and open-source code for training commercial LLMs.

IEEE-CS / ACM Software Engineering 8 Principles

01. PUBLIC (TOP PRIORITY)

Public Interest First

Act consistently with public health, safety, and welfare.

02. CLIENT & EMPLOYER

Aligned Interests

Serve clients loyally while remaining compliant with public good.

03. PRODUCT

Quality Standards

Ensure highest professional standards and rigorous QA.

04. JUDGMENT

Independence

Maintain professional integrity and impartial assessment.

05. MANAGEMENT

Ethical Leadership

Promote ethical software development workflows.

06. PROFESSION

Reputation

Advance the integrity of software engineering discipline.

07. COLLEAGUES

Fair Support

Be fair to and supportive of professional colleagues.

08. SELF

Lifelong Learning

Continuously improve skills in ethical engineering practice.

Unethical Risks vs. Ethical Practice Standards

Domain	Unethical Practice	Ethical Engineering Standard
UX & Choice Architecture	Deceptive dark patterns & forced retention flows	Transparent, reversible, user-centric choices
User Data & Telemetry	Uninformed harvesting & re-identifying records	Privacy-by-design & minimal data collection
AI & Model Training	Unverified training data with historical bias	Rigorous demographic fairness auditing
Security & Release QA	Deploying known vulnerabilities to meet deadlines	Pre-deployment safety gates & continuous patching

Essential Literature in Software Ethics

Cathy O'Neil

Weapons of Math Destruction

Exposing opaque, unregulated mathematical models driving inequality.

Ruha Benjamin

Race After Technology

Analyzing how systemic racial bias gets encoded into "neutral" tech.

Virginia Eubanks

Automating Inequality

Investigating automated decision profiling in public welfare systems.

Safiya Noble

Algorithms of Oppression

Structural discrimination embedded in commercial search engines.

Kearns & Roth

The Ethical Algorithm

Formal mathematical techniques for embedding privacy & fairness.

Brian Christian

The Alignment Problem

How machine learning models learn human values and misalignments.



Ethics is a Design Constraint, Not an Afterthought.

Real-world failures trace back to treating ethics as an afterthought. Ask at design time: *"Who could this harm, and how would we know?"*